1(a)(i)

average thickness = 3.8 / 20 = 0.19 cm

(b)

density = mass / volume = 148 / 16.6 = 8.9 g/cm3

4(a)

speed = distance / time = 200 / 6.4 = 31 m/s

5(b)

clockwise moments = anticlockwise moments. 200 = (2.0 x 10) + (F x 60). F =

(200-20) / 60 = 180 / 60 = 3.0 N

11(b)

Vs/Vp = Ns/Np, so Vs = (64 / 240) x 960 = 16 V

12(b)

137 - 56 = 81

(c)

three half-lives have passed, so 36 / 8 = 4.5 mg